CCEE Big Data Technologies: Study Cheatsheet

Session 1 - 3 Summary

1. Big Data Fundamentals

Definition

Big Data refers to datasets whose size or type is beyond the ability of traditional relational databases to capture, manage, and process with low latency.

The 5 V's (Memorize these)

- **Volume:** The scale of data (Terabytes, Petabytes).
- **Velocity:** Analysis of streaming data (Speed of generation and processing).
- **Variety:** Different forms (Structured, Unstructured, Semi-structured).
- **Veracity:** Uncertainty of data (Quality, Accuracy).
- **Value:** Business value derived from data.

EXAM ALERT: Do not confuse Veracity (Quality) with Variety (Format).

The Fundamental Shift

- **RDBMS (SQL):** Uses **Schema-on-Write**. You must define the table structure BEFORE inserting data. Good for consistency.
- **Big Data (Hadoop):** Uses **Schema-on-Read**. You dump raw data now and define the structure ONLY when you query it. Good for flexibility.

2. Data Repositories & Lifecycle

Repository Types

- **Data Warehouse:** Stores highly structured, historical data. (Expensive, strict).
- **Data Lake:** Stores raw data in native format (Structured + Unstructured). Low cost.
- **Meta-Repository:** A catalog/directory describing other repositories. It helps researchers find WHERE data is located (like a phonebook).

Data Curation Concepts

- **Data Curation:** Active management of data over its lifecycle (tagging, archiving).
- **FAIR Principles:** Findable, Accessible, Interoperable, Reusable.
- **Data Rescue:** Recovering data from obsolete formats or decaying media (e.g., reading data from 1990s floppy disks before they rot).
- **OpenExit:** A strategy to ensure you can move your data OUT of a cloud provider/repository if they shut down (prevents Vendor Lock-in).

3. Hadoop History & Core Components

Origins

- **Founders:** Doug Cutting and Mike Cafarella.
- **Project:** Originally part of 'Nutch' (search engine).
- **Inspiration:** Google's whitepapers on **GFS** (Google File System) and **MapReduce**.
- **Name:** Named after Doug Cutting's son's yellow toy elephant.

The Core 3 Components (Hadoop 2.0+)

- **HDFS (Hadoop Distributed File System):** The **Storage** Layer. It breaks files into Blocks and replicates them.
- **YARN (Yet Another Resource Negotiator):** The **Management** Layer. It allocates RAM and CPU. It acts as the 'Operating System' of the cluster.
- **MapReduce:** The **Processing** Layer. Logic for processing data (Map = Sort/Filter, Reduce = Summarize).

EXAM ALERT: YARN manages Resources. MapReduce processes Data. HDFS stores Data.

4. HDFS Deep Dive

Block Storage

- Files are split into chunks called 'Blocks'.
- **Default Block Size (Hadoop 2.x/3.x):** 128 MB.
- **Default Replication Factor:** 3 (Data is stored on 3 different nodes).

Daemons (Processes)

- **NameNode (Master):** Stores Metadata (File names, permissions, location of blocks). It does **NOT** store actual file contents.
- **DataNode (Slave):** Stores the actual data blocks.

EXAM ALERT: If the NameNode crashes (and has no backup), the entire cluster is useless because you lose the file mapping, even if the DataNodes are fine.

5. Hadoop Version Evolution

Hadoop 1.x

- Single Point of Failure (NameNode).
- **JobTracker** handled BOTH resource management and job scheduling (Bottleneck).
- **Limitation:** Could ONLY run MapReduce jobs.

Hadoop 2.x (The Revolution)

- Introduced **YARN**.
- Split JobTracker into **ResourceManager** and **ApplicationMaster**.
- **Benefit:** Can run Spark, Storm, and MapReduce simultaneously.
- **High Availability:** Supported Standby NameNodes.

Hadoop 3.x

- **Erasure Coding:** Reduces storage overhead by $\sim$50% compared to 3x replication (Saves disk space).
- **Multiple NameNodes:** Supports 2+ NameNodes.
- **GPU Support:** For Deep Learning.

EXAM ALERT: Erasure coding saves SPACE, it does not necessarily increase processing SPEED.

6. Distributions

- **Apache Hadoop:** Pure open source, hard to configure.
- **Cloudera (CDH):** Market leader, proprietary management tools.
- **Hortonworks (HDP):** Was 100% open source.
- **Note:** Cloudera and Hortonworks merged in 2019.